A NEW PROOF OF THE H-COLOURING DICHOTOMY

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Abstract. In this note, we present a new proof of the H-colouring Dichotomy, which was proved by Hell and Nešetřil in 1990, and was reproved by Bulatov in 2005. The proof is much shorter than the original proof, and avoids the algebraic machinery of Bulatov's proof.

1. Introduction

Arguably, one of the most celebrated results in graph homomorphism theory is the H-colouring Dichotomy, or Graph Homomorphism Dichotomy, of Hell and Nešetřil, [3].

A *homomorphism of a graph G to a graph H*, or an H-colouring of G, is a mapping ϕ of the vertices of G to the vertices of H such that $\phi(u)\phi(v)$ is an edge of H if uv is an edge of G. The problem H-COL is a computational problem in which one is given an instance, a graph G, and one must determine if there exists an H-colouring of G. The question of interest for a given H, is what is the computational complexity of the problem H-COL, in terms of the size of G.

The following result of Hell and Nešetřil, stated here for loopless graphs, is known as the H-colouring Dichotomy.

Theorem 1.1. [3] *The problem H-COL is polynomial time solvable if H is bipartite, and is NP-complete otherwise.*

The proof of Hell and Nešetřil uses purely graph theoretic techniques, but is fairly long and very intricate. In [1], Bulatov used techniques from Universal Algebra, popularised in [2, 5], to simplify the proof of Hell and Nešetřil. His reproof was a byproduct of showing that the H-colouring Dichotomy agreed with the more general CSP Dichotomy Classification conjectured in [2].

In [9, 10], a construction called the 'fibre construction' was developed, and was used to give low machinery proof of many relevant results previously proved by algebraic means. In particular, a graph theoretic interpretation of the CSP Dichotomy Classification Conjecture was given.

In this note, we reprove the H-colouring Dichotomy using a variation of the fibre construction from [10]; or more accurately, we prove Proposition 3.1, which replaces the hardest part of the original proof.

This is a clarification of the published version, which appears in SIAM J. Discrete Math. Volume 23, Issue 4, pp. 2204-2210 (2010), available at http://dx.doi.org/10.1137/080736697. The published version has several typos, (which I introduced after the paper was refereed,) mostly concerning the notation $p_1(G)$ (and $p_2(G)$) which is defined as one more than the number of vertex (resp. vertex pair) orbits of G, but is often treated as the number of orbits. In this version, I have corrected these mistakes, and further, have seperated results unnecessary to the proof of the dichotomy, from the main part of the paper.

Our new proof retains important ideas introduced in both of the earlier proofs, but is shorter than the proof of [3] and does not require the algebraic machinery used in [1]. As well as being a streamlined graph theoretic proof of Theorem 1.1, our proof gives a solid intuition as to why certain reductions are natural. Furthermore, we view this proof as further evidence that the fibre construction of [10] will be a useful tool in the study of CSPs.

2. Notation and definitions

All graphs are simple, loopless and undirected. We denote an edge containing the vertices u and v by uv, and denote the fact that it is an edge by $u \sim v$. An H-colouring ϕ of a graph G is an *injection* if $\phi(u) = \phi(v)$ implies that $u = v$. An *automorphism of* H is an injection from H to H. We write $G \to H$ if there exists an H-colouring of G. We denote the fact that S is a subgraph of G by $S \leq G$. A graph C is a *core* if its only C-colourings are automorphisms. It is well known (see [4]), and easy to show, that every graph H has a unique (up to isomorphism) subgraph H_C which is a core. Furthermore $H \to H_C$ and $H_C \to H$, so H-COL has the same computational complexity as H_C-COL.

The complete graph K_3 will always have vertex set $\{1, 2, 3\}$. When we denote a function f acting on a vector $(v_1, \ldots, v_d)$ we drop a set of parentheses and write $f(v_1, \ldots, v_d)$.

3. From Theorem 1.1 to Proposition 3.1

As K_2 (or K_1) is the core of every bipartite graph, it is easy to see that H-COL is polynomial time solvable for bipartite graphs. Indeed, K_2-COL is equivalent to checking if the instance G is bipartite, and K_1-COL is equivalent to checking for the existence of edges.

The hard part of Theorem 1.1 is showing that H-COL is NP-complete if H contains an odd cycle. Hell and Nešetřil's first step in proving this was reducing it to the case in which H has the following properties:

- H is a core.
- Every vertex in H is in a triangle.
- H contains no copies of K_4.

These reductions are standard, and Bulatov follows the proof exactly up to this point, but here the proofs diverge. However, with different proofs, they both go on to further reduce to the following case:

- Every edge of H lies in a unique triangle.

The details of these reductions can be found in any number of places, among others, [3, 4] and [1], and we add nothing to them. We thus pick up the proof from here, that is, we prove the following.

Proposition 3.1. *H-COL is NP-complete for any core graph H, with at least one edge, in which every edge lies in a unique triangle.*

The proof of Proposition is immediate from Lemmas 4.2 and 5.1, which we prove in Sections 4 and 5 respectively. Section 6 contains a generalisation of the ideas in Section 4.

4. A Lemma about Projectivity

Definition 4.1. For an integer $d \geq 1$ the dth power K_3^d of K_3 is the graph with vertex set

$$\{(x_1, \ldots, x_d) \mid x_1, \ldots x_d \in \{1, 2, 3\}\},$$

and in which $(x_1, \ldots, x_d) \sim (x'_1, \ldots, x'_2)$ if and only if $x_i \neq x'_i$ for $i = 1, \ldots, d$.

A homomorphism $\phi : K_3^d \to H$ is called projective if it *projects* onto some for some non-empty *index set* $I \subseteq [d]$; that is, if

$$\phi(x_1, \ldots, x_d) = \phi(x'_1, \ldots, x'_d)$$

if and only if $x_i = x'_i$ for all $i \in I$. In particular, if ϕ is an injective homomorphism, it is a projection.

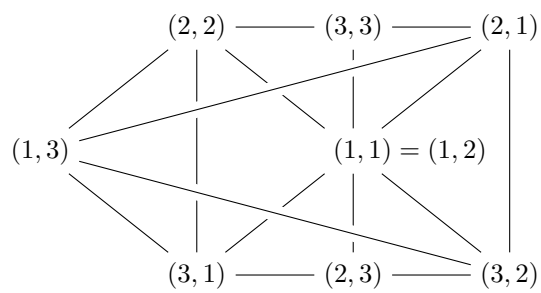
By the symmetry of K_3^d, it is clear that any map $\sigma : K_3^d \to K_3^d$ which permutes coordinates, or permutes the entries in any coordinate, or is the composition of such operations, is an automorphism of K_3^d. Further, it is clear that an H-colouring ϕ of K_3^d is projective if and only if $\phi \circ \sigma$ is for any such automorphism σ.

The following lemma is implicit in Bulatov's reproof of the H-colouring Dichotomy. However, where his proof uses the algebraic machinery developed in [2], we give a simple proof. By Lemma 6.4 it is enough to prove in the case that $d = 2, 3$, and this is what we did in the published version of the paper. However, it is clearer to prove it directly here.

Lemma 4.2. *Let H be a graph in which no edge occurs in more than one triangle. Then all H-colourings of K_3^d are projective, for all $d \geq 2$.*

Proof. We first prove the case $d = 2$, and then proceed by induction on d.

Case: $d = 2$**.** Let ϕ be an H-colouring of K_3^2. If ϕ is an injection, we are done, so by symmetry, we may assume that ϕ maps $(1, 1)$ and $(1, 2)$ to the same image. The following figure shows K_3^2 in which these vertices have been identified.



Observe that any two vertices that have the same entry in the first coordinate are non-adjacent vertices occurring in distinct triangles with a common edge, thus must be mapped to the same image in H. Identifying such pairs of vertices, we find that ϕ is a projection onto the first coordinate.

Now assume that $d \geq 3$ and that any homomorphism $K_3^c \to H$ $2 \leq c < d$ is projective. Let ϕ be an H-colouring of K_3^d. If ϕ is injective then we are done, so up to symmetry we may assume that $\phi(1, 1, \ldots, 1) = \phi(1, \ldots 1, 2, \ldots, 2)$ where in

the second vertex, there are r ones. We will show that ϕ is independent of the last coordinate; that is, that for any choice of entries,

$$\phi(x_1, \ldots, x_d) = \phi(x_1, \ldots, x_{d-1}, x'_d) \tag{1}$$

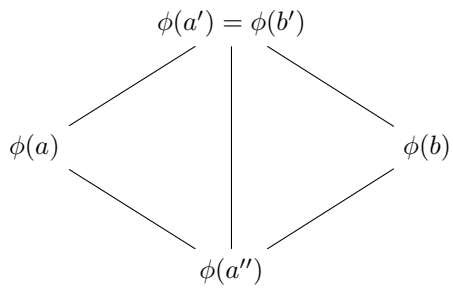
Case: $r > 1$**.** First we show that (1) holds when $x_1 = x_2$. Let $\phi'(x_2, x_3, \ldots, x_d) = \phi(x_2, x_2, \ldots, x_d)$. As ϕ' is projective by induction, and

$$\phi'(1, \ldots, 1) = \phi(1, \ldots, 1) = \phi(\underbrace{1, \ldots, 1}_{r}, 2, \ldots, 2) = \phi'(\underbrace{1, \ldots, 1}_{r-1}, 2, \ldots, 2),$$

ϕ' projects onto an index set not containing the last index. Thus

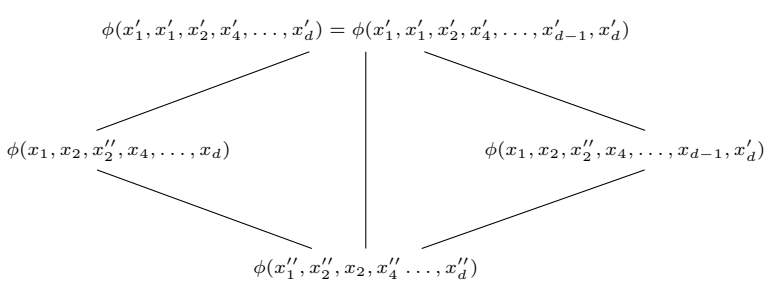
$$\phi(x_2, x_2, \ldots, x_d) = \phi'(x_2, \ldots, x_d) = \phi'(x_2, \ldots, x_{d-1}, x'_d) = \phi(x_2, x_2, \ldots, x_{d-1}, x'_d)$$

and so (1) holds when $x_1 = x_2$. We now show that also (1) holds when $x_1 \neq x_2$. For any i let x_i, x'_i, and x''_i be distinct vertices of K_3. So we may write $x_1 = x''_2$. Then $a = (x''_2, x_2, x_3, \ldots, x_d)$, $a' = (x'_2, x'_2, x'_3, \ldots, x'_d)$ and $a'' = (x_2, x''_2, x''_3, \ldots, x''_d)$ are a triangle, and a'', $b = (x''_2, x_2, x_3, \ldots, x_{d-1}, x'_d)$, and $b' = (x'_2, x'_2, x'_3, \ldots, x'_{d-1}, x''_d)$ are a path. The same is true after applying ϕ, but $\phi(a') = \phi(b')$, so have the following graph in the image of ϕ.



But this graph doesn't exist in H, so $\phi(a) = \phi(b)$, which is (1) (because $x_1 = x''_2$).

Case: $r = 1$**.** The proof is similar to the case $r > 1$. Using $\phi'(x_1, x_2, x_4, \ldots, x_d) = \phi(x_1, x_2, x_2, \ldots, x_d)$ we get that (1) holds in the case that $x_2 = x_3$. In the case that $x_2 \neq x_3 = x''_2$, we have the following graph in the image of ϕ:



In the same way as in the previous case, this shows that (1) holds.

As (1) holds, $\phi(x_1, \ldots, x_d) = \phi'(x_1, \ldots, x_{d-1})$ for some H-colouring ϕ' of K_3^{d-1}, which is projective by induction. So ϕ is projective. This completes the proof of the lemma. $\square$

5. The Fibre Construction

We prove the following lemma using a version of the Fibre Construction from [10], which we have tailored to the present problem.

Lemma 5.1. *If H is a core containing a copy of K_3 such that any H-colouring of K_3^d is projective for all $d \geq 2$, then H-COL is NP-complete.*

Proof. We present a polynomial time construction which provides, for an instance G of K_3-COL, an instance $M(G)$ of H-COL, such that $G \to K_3 \iff M(G) \to H$. As K_3-COL is NP-complete [6], this will show that H-COL is NP-complete.

Let d be the maximum integer such that H contains a copy of K_3^d, and let T be a fixed copy of K_3^d in H. Let the vertices of T be d-tuples $\dot{x} = (x_1, \ldots, x_d)$ of the set $\{1, 2, 3\}$, such that $\dot{x} \sim \dot{y}$ if and only if all coordinates are different.

In the construction, copies of the graph $\mathscr{T} \cong (K_3^d)^6$, will figure prominently. Vertices of $\mathscr{T}$ are 6-tuples of d-tuples of the set $\{1, 2, 3\}$. We give special labels to the following two vertices:

$$\ddot{a} = ((1, \ldots, 1), (1, \ldots, 1), (2, \ldots, 2), (2, \ldots, 2), (3, \ldots, 3), (3, \ldots, 3)),$$
$$\ddot{b} = ((2, \ldots, 2), (3, \ldots, 3), (1, \ldots, 1), (3, \ldots, 3), (1, \ldots, 1), (2, \ldots, 2)).$$

Given a copy $\mathscr{T}_e$ of $\mathscr{T}$, the copy of a vertex $(\dot{x}_1, \ldots, \dot{x}_6)$ is labelled $(\dot{x}_1, \ldots, \dot{x}_6)_e$ and in particular, the copies of $\ddot{a}$ and $\ddot{b}$ are labelled $\ddot{a}_e$ and $\ddot{b}_e$.

Construction 5.2. *Given a graph G, arbitrarily orient its edges. For every arc e of G let $\mathscr{T}_e$ be a copy of $\mathscr{T}$. Build $M(G)$ from G, H and $\mathscr{T}_e$ for each $e \in E(G)$, as follows.*

(i) *For each arc $e = uv \in E(G)$, identify u and v of G with $\ddot{a}_e$ and $\ddot{b}_e$ respectively.*
(ii) *For each $e \in E(G)$ and each vertex $\dot{x} \in V(T)$, identify $(\dot{x}, \ldots, \dot{x})_e$ in $\mathscr{T}_e$ with $\dot{x}$ in $T \leq H \leq M(G)$.*

We show that $G \to K_3 \iff M(G) \to H$. Let ϕ be an H-colouring of $M(G)$. As $T \cong K_3^d$ is K_3-colourable, the following claim implies that G is K_3-colourable, giving us the implication $M(G) \to H \Rightarrow G \to K_3$.

Claim 5.3. *The homomorphism ϕ maps $G \leq M(G)$ to a subgraph of H isomorphic to T.*

Proof of claim. Let uv be an arc of G. Since ϕ maps $H \leq M(G)$ to H and H is a core, we may assume that ϕ is the identity on H. By the identification of $(\dot{x}, \ldots, \dot{x})_{uv} \in \mathscr{T}_{uv}$ with $\dot{x} \in T \leq H \leq M(G)$ we have that $V(T) \subseteq \phi(\mathscr{T}_{uv})$. Moreover since $\mathscr{T}_{uv}$ is isomorphic to K_3^{6d}, so by the premise of the lemma maps to a power of K_3 in H, and T is contained in no greater power of K_3 in H, ϕ maps $\mathscr{T}_{uv}$ onto T. In particular ϕ maps the edge $\ddot{a}_{uv}\ddot{b}_{uv}$ of $\mathscr{T}_{uv}$ to an edge of T. But u and v are identified with $\ddot{a}_{uv}$ and $\ddot{b}_{uv}$ respectively, so ϕ maps uv to an edge of T. $\diamond$

To see the implication $G \to K_3 \Rightarrow M(G) \to H$, we let ϕ be a K_3-colouring of G, and define an H-colouring ϕ' of $M(G)$ as follows. Let ϕ' be the identity on $H \leq M(G)$. Extend ϕ' to $G \leq M(G)$ by letting $\phi'(v)$, for $v \in V(G)$, be the d-tuple $(\phi(v), \ldots, \phi(v))$ in $T \leq H$. We have just to verify that ϕ' can be extended to an H-colouring of $\mathscr{T}_e$ for each $e = uv \in E(G)$. The only vertices of $\mathscr{T}_e$ on which ϕ'

has already been defined, are $\ddot{a}_e$ and $\ddot{b}_e$, and $(\dot{x},\dots,\dot{x})_e$ for each $\dot{x}\in T$. Indeed, $\phi'(\ddot{a}_e)=(\phi(u),\dots,\phi(u))$ and $\phi'(\ddot{b}_e)=(\phi(v),\dots,\phi(v))$ are d-tuples in T, where $\phi(v)$ and $\phi(u)$ are different elements of $i\in\{1,2,3\}$, and $\phi'((\dot{x},\dots,\dot{x})_e)=\dot{x}\in T\le H$ for $\dot{x}\in T$. Recalling the definition of the vectors $\ddot{a}$ and $\ddot{b}$, one sees that these mappings are consistent with a projection of $\mathscr{T}_e$ onto one of its six coordinates. We may thus extend ϕ' to a T-colouring of $\mathscr{T}_e$ by extending it to this projection. Doing this for all $\mathscr{T}_e$ we extend ϕ' to an H-colouring of $M(G)$, as needed. □

6. More Projectivity

The proof of the dichotomy is complete. This section expands on the ideas from Section 4, and is of independent interest.

Definition 6.1. Given a graph H and an integer $d\ge 1$ the dth *(categorical) power* H^d *of* H is defined as the graph whose vertex set is the set $V(H)^d$ of d-element vectors over $V(H)$, and whose edges are defined by

$$(u_1,\dots,u_d)\sim(v_1,\dots,v_d)\iff u_i\sim v_i \text{ for all } i=1,\dots,d.$$

An H-colouring ϕ of H^d is called a (d-ary) *polymorphism* of H.

It is easy to check that for any $i\le d$, the projection π_i of H^d defined for all vectors $\dot{v}=(v_1,\dots,v_d)\in V(H^d)$ by $\pi_i(\dot{v})=v_i$, is a polymorphism. A graph H is called *projective* if its only polymorphisms are projections composed with automorphisms of H. In [5], Jeavons showed that if H is a projective graph (any projective relational structure, in fact), then H-COL is NP-complete.

In [8] it was shown for any connected graph G with at least 3 vertices, that if all 2-ary polymorphisms of G are projections, then all polymorphisms are projections. We now look at the same concept, but for homomorphisms of powers of G to an arbitrary target H, rather than just to G.

Definition 6.2. For any $I=\{\alpha_1,\dots,\alpha_{|I|}\}\subseteq\{1,2,\dots,d\}$ with $\alpha_i<\alpha_{i+1}$ for all i, define the *(generalised) projection* $\pi_I:H^d\to H^{|I|}$, by letting

$$\pi_I(\dot{v})=(v_{\alpha_1},\dots,v_{\alpha_{|I|}})$$

for all vectors $\dot{v}=(v_1,\dots,v_d)\in V(H^d)$.

It is again easy to check that this is a homomorphism.

Definition 6.3. An H-colouring ϕ of G^d is called *projective* if $\phi=\imath\circ\pi_I$ for some projection $\pi_I:G^d\to G^{|I|}$, with $|I|\le d$, and some injection $\imath:G^{|I|}\to H$.

Given a graph G, let $p_1(G)$ denote the number of orbits of vertices, under the group of automorphisms of G and let $p_2(G)$ denote the number of orbits of ordered pairs of vertices. Thus $p_1(K_k)=1$, $p_2(K_k)=2$ and for any G, $p_1^2(G)\le p_2(G)\le|V(G)|^2$.

Lemma 6.4. *For any graphs G and H if all H-colourings of $G^{p_2(G)+1}$ are projective then all H-colourings of any power of G are projective.*

Proof. We show by induction on n that any H-colouring ϕ of G^n is projective. All H-colourings of $G^{p_2(G)+1}$ being projective clearly implies all H-colourings of G^d are projective for $d<p_2(G)+1$, so we may assume that $n\ge p_2(G)+2$.

If ϕ is independent of any coordinate, then it factors through an H-colouring of G^{n-1}, so is projective by induction. Thus we may assume that it depends on every coordinate.

Claim 6.5. *For any choice of $1 < i < j \leq n$, and any automorphism τ of G, there exist vertices $v_1, \ldots, v_n$ and $v_1' \neq v_1$, such that $v_j = \tau(v_i)$ and $\phi(v_1, v_2, \ldots, v_n) \neq \phi(v_1', v_2, \ldots, v_n)$.*

Proof of claim. As ϕ depends on the first coordinate, there are vertices $u_1, \ldots, u_n$ and $u_1' \neq u_1$ such that $\phi(u_1, u_2, \ldots, u_n) \neq \phi(u_1', u_2, \ldots, u_n)$. As $n - 2 \geq p_2(G) > p_1(G)$ there are some $1 < s < t \leq n$, $s \neq i$ such that $u_t = \sigma(u_s)$ for some automorphism σ of G. Define an H-colouring ϕ' of G^{n-1} by

$$\phi'(x_1, \ldots, x_{n-1}) = \phi(x_1, \ldots, x_{t-1}, \sigma(x_s), x_t \ldots, x_{n-1}).$$

By induction, ϕ' is projective, and because

$$\begin{aligned} \phi'(u_1, u_2, \ldots, u_{t-1}, u_{t+1}, \ldots, u_n) &= \phi(u_1, u_2, \ldots, u_n) \\ &\neq \phi(u_1', u_2, \ldots, u_n) \\ &= \phi'(u_1', u_2, \ldots, u_{t-1}, u_{t+1}, \ldots, u_n), \end{aligned}$$

it projects onto some index set I containing 1. Thus $\phi(x_1, \ldots, x_n) \neq \phi(x_1', \ldots, x_n')$ whenever $x_t = \sigma(x_s)$, $x_t' = \sigma(x_s')$, and $x_1 \neq x_1'$. In particular $\phi(v_1, \ldots, v_n) \neq \phi(v_1', \ldots, v_n)$ for any choice of $v_1, \ldots, v_n$ and $v_1' \neq v_1$ such that $v_j = \tau(v_i)$ and $v_t = \sigma(v_s)$. As $i \neq s$, there is some such choice. $\diamond$

Claim 6.6. *Whenever $x_1 \neq x_1'$, $\phi(x_1, \ldots, x_n) \neq \phi(x_1', \ldots, x_n')$.*

Proof of claim. Because $n-1 \geq p_2(G)$, there exist some $1 < i < j \leq n$, and some automorphism σ of G, such that $x_j = \sigma(x_i)$ and $x_j' = \sigma(x_i')$. By the previous claim, there exist $v_1, \ldots, v_n$ and $v_1' \neq v_1$, such that $v_j = \sigma(v_i)$ and $\phi(v_1, v_2, \ldots, v_n) \neq \phi(v_1', v_2, \ldots, v_n)$. Using this fact, we can show, just like in the previous claim, that the H-colouring ϕ'' of G^{n-1} defined

$$\phi''(x_1, \ldots, x_{n-1}) = \phi(x_1, \ldots, x_{j-1}, \sigma(x_i), x_j \ldots, x_{n-1}),$$

projects onto some index set I containing 1. Thus $\phi(x_1, \ldots, x_n) \neq \phi(x_1', \ldots, x_n')$. $\diamond$

By symmetry, claims similar to Claim 6.6 are true for any index in place of 1. Thus ϕ is an injection, and so projective. $\square$

This upper bound on the degree we have to check to decide if there are non-projective homomorphisms from powers of G to H is not tight. In particular, for any clique K_k we have that $p_2(K_k) + 1 = 3$, but in Lemma 4.2 we showed that if all H-colourings of K_k^2 are projective, then all H-colourings of K_k^d are projective for any d.

It would be interesting to find exactly what degree one has to check for a given graph G.

7. Remarks and Acknowledgments

• Lemma 5.1 could be proved for all graphs P in place of K_3, and could be proved more quickly by showing that H is K_3-*partitionable*, as defined in [10]. However we prove it directly to avoid the heavier definitions of this paper. See below for details.

• A new, completely different proof of the H-colouring Dichotomy is given in a recent paper of Kun and Szegedy [7].
• I would like to express my gratitude to Gábor Tardos for many suggestions which improved the presentation. More importantly, I thank him for his help in improving the proof of Lemma 6.4 and for his observation that it can be proved for all graphs, rather than just cliques. I would also like to thank Pavol Hell for urging me to publish this, and him, Andrei Bulatov, and Jarik Nešetřil for offering useful comments.
• I would like to thank the referee's of the original published paper for their helpful comments.

8. Added 2014

I have been asked about the comment that Lemma 5.1 could be proved more quickly by showing that H is K_3-*partitionable*. Here are some details.

In [10] we give the following definition.

Definition 8.1. A structure H as K_3-*partitionable* if there is

- an instance M of H-COL containing disjoint ordered sets W^a and W^b of d vertices each,
- disjoint families $\mathcal{P}_1$, $\mathcal{P}_2$, $\mathcal{P}_3$ of ordered d-tuples of $V(H)$ (so-called *patterns*),
- such that any homomorphism $\phi : M \to H$ restricts on W^a and W^b to patterns in distinct families $\mathcal{P}_i$ and $\mathcal{P}_j$, and
- for any choice of distinct $i, j \in \{1, 2, 3\}$ there is a homomorphism $\phi : M \to H$ that retricts on W^a to a pattern in $\mathcal{P}_i$ and on W^b to a pattern in $\mathcal{P}_j$.

In [10] we show that if H has a k_3-partition, then one can reduce K_3-COL to H-COL, showing the latter is NP-complete. Lemma 5.1 could be proved by showing that H is K_3-partitionable, as follows.

Proof. Where d is the greatest integer such that H contains a copy of K_3^d, and T is a fixed copy of K_3^d in H, G be constructed from the graph $\mathscr{T} = (K_3^d)^6$ and a copy of H_0 by identifying the diagonal copy of T in $\mathscr{T}$ (the one on the vertices $(\dot{x}, \ldots, \dot{x})$) with $T \leq H_0$.

We show that G is a K_3-partition of H. The singletons $\{\ddot{a}\}$ and $\{\ddot{b}\}$ are sets W^a and W^b respectively, and $\mathcal{P}_i$ is the family of d-tuples whose first element is i.

Any homomorphism $\phi : G \to H$ maps the core H_0 identically (we may assume,) to H, so maps the diagonal copy of T in $\mathscr{T}$ to $T \leq H$. The rest of $\mathscr{T} = (K_3^d)^6$ must map projectively onto T, so ϕ restricted to $\mathscr{T}$ is a projection onto d slots. Identification of the diagonal T in $\mathscr{T}$ with T in H ensures that these slots must have indices $i_1, \ldots i_6$ where i_j is some number congruent to j modulo 6. Whichever the first of these slots is, ϕ maps $\ddot{a}$ and $\ddot{b}$ to patterns in different families $\mathcal{P}_i$ and $\mathcal{P}_j$.

That such a map exists for any choice of i and j is trivial.

□

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